

SmartHire : Intelligent Resume Shortlisting System

**A Project Report
Submitted
In Partial Fulfillment of the Requirements
For the Degree of
Bachelor of Technology (B.Tech)
in
Computer Science Engineering(Data Science)
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Declaration

We hereby declare that the project work presented in this report entitled “**SmartHire : Intelligent Resume Shortlisting System**”, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering (Data Science), submitted to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh , Lucknow is based on our own work carried out at the Department of Computer Science & Engineering(Data Science), G.L. Bajaj Institute of Technology & Management, Greater Noida. The work contained in the report is true and original to the best of our knowledge and project work reported in this report has not been submitted by us for award of any other degree or diploma.

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Abstract

Manual screening of resumes has become a slow, inconsistent, and unreliable process in current recruitment practices. With huge volume of applications, it becomes necessary for the software to pre-screen the resume before a human recruiter processes it. Most of the existing systems adopt a filtering approach that uses keyword search in resumes and sometimes in job descriptions. The same set of skills can be worded differently in resume, therefore, the current systems lack the speed, accuracy, and most importantly, intelligibility. The recruiter needs to know why a particular resume has been shortlisted for the next round of interview, and why another one has been rejected. This project proposes SmartHire: Intelligent Resume Shortlisting System, a web-based application that automates the initial screening of resumes for a recruiter. The system is designed and developed using Python programming language and built on top of Flask framework.

We can accept in PDF/DOCX format and process the resume and job description in-house using Natural Language Processing. We extract the relevant information from the resume such as skills and contact details. We then use a hybrid scoring method that is based on semantic similarity (sentence-level embeddings) and matching of keywords in the resume and job description. We are able to adjust the weightings of both scores based on the job role ensuring that we are not missing out on key responsibilities that are described in the job description using keywords.

This API not only allows users and recruiters to check Resume vs Job Description to get a score along with details of matched skills and missing skills but also allows you to upload bulk resumes and rank candidates based on extracted contact information. The API also evaluates performance using precision, recall, accuracy and F1-score on labeled data.

The proposed system is efficient in reducing the manual work, greatly increasing the speed of screening and providing clear results with substantial explanation. The system has the potential to reduce the manual work and increase the screening speed for resume shortlisting, and it can be applied to actual recruitment practice and academic research using AI technique combined with high practicability.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ATS	Applicant Tracking System
TF-IDF	Term Frequency – Inverse Document Frequency
OCR	Optical Character Recognition
IR	Information Retrieval
LLM	Large Language Model

Chapter 1

Introduction

1.1 Preliminaries

In recent times, there have been some changes in the recruitment process, thanks to the advancements made in the area of digitalization. There have been an increase in the number of resumes received for each job post as compared to previous recruiting methods used. Despite being a good thing in increasing the available pool of applicants, this has become a great challenge for recruiters as it is hard to filter the right candidates among such a huge number of applicants in a short period. Hence, resume screening becomes a necessity before advancing to other phases like conducting interviews.

The definition of resume screening basically entails comparing a candidate's qualifications to the ones required in the job post. The process involves analyzing each resume separately. However, this task is very challenging for organizations due to the huge number of resumes received on each job posting. Furthermore, the screening process is done manually; thus, making it time-consuming and also inconsistent depending on the recruiter doing the analysis. Therefore, there have been attempts of coming up with automation tools to speed the process. The process of intelligent resume screening usually comprises three primary steps, including data extraction from the document, encoding or structuring of this information, and finally computing the degree of similarity between the resume content and job description. These stages are typical for most intelligent screening applications that are currently developed by modern researchers.

One of the core ideas employed in such intelligent systems is semantic similarity. It is an alternative approach to keyword-based search that is focused on interpreting the meaning of a certain text rather than matching particular words. In other words, phrases such as software developer and programmer may denote almost identical positions, yet their keywords will not match each other. To calculate semantic similarity, there are a number of methods employed,

including sentence embedding. With its help, a document is represented numerically taking into account its semantics.

While semantic analysis is crucial, it is equally necessary to have a component that matches the candidate based on keywords. Some of the requirements in a job, for example, certain programming languages or applications, must be included in a candidate’s profile. This component makes sure that the most crucial requirements in a job are met through keyword matching. The use of this technology has been improved by including domain knowledge and synonyms to increase the flexibility of the search process.

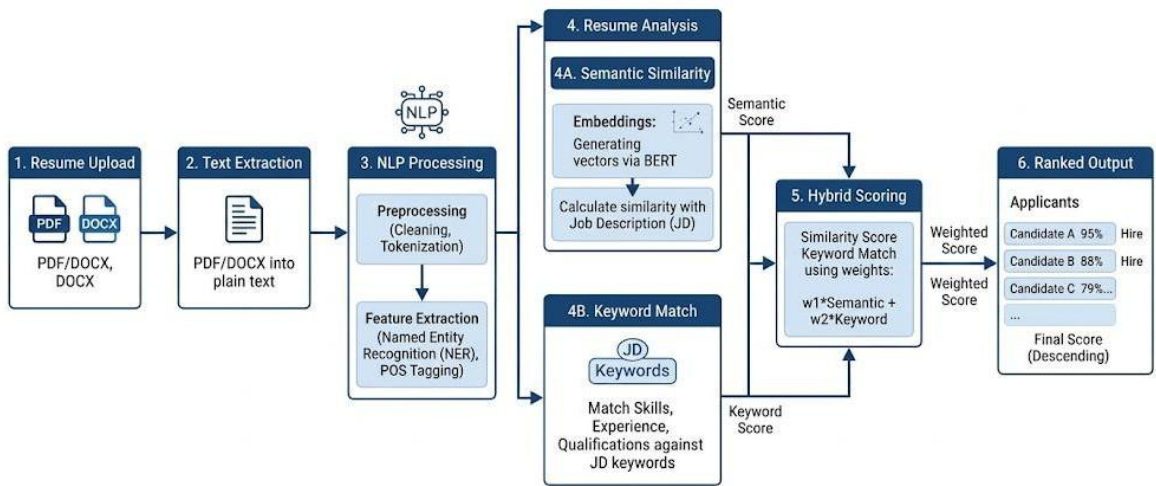


Fig. 1.1 Intelligent Resume Screening System Workflow

By combining semantic similarity analysis and keyword matching, a hybrid method results, which is generally believed to be better than using just one of the methods. Semantic analysis gives a more holistic idea of the candidate, while keyword matching guarantees that all requirements for the position are fulfilled. Assigning weights to these two processes allows generating a score that indicates how suitable the candidate is for a particular position.

The ability of the system to handle Natural Language (NL) commands and answers (if required) is another feature of the system that could enhance the experience of the users. The system is able to parse and understand commands written in Natural Language and it is also able to respond to these commands in Natural Language as well. This ability of the system to handle NL is another feature of the system.

Artificial Intelligence. The system uses AI (NLP) to automatically analyze messages. The system

can understand human language.

These are words extracted from a resume and a job description. The task involves text preprocessing, tokenization, etc.

Extracting skills using tools like spaCy and NLTK.

The main goal of NLP is to convert unstructured language data to structured computational language.

Natural Language Processing (NLP) is utilized in the intelligent resume screening process.

It incorporates NLP functions. Optionally, it can use the capacity of Large Language Models.

Models (LLM), for example Google Gemini. The main advantage of LLMs is that they are very fast. They are pre-trained on an enormous amount of data and have a WordPiece tokenization similar to BERT.

Job Description Generator tool that generates job description, listing required skills and briefly describes job responsibilities.

This system gives feedback to the user about his/her qualifications. In this project I didn't pay attention to give feedback to the user about his/her psychological type.

This large language model is intended to be an optional function.

System evaluation procedures—meant to measure system efficiency during design—are included in the manual. Procedures for evaluating the efficiency of the current system are created in order to later compare these results with results acquired in the evaluation of the proposed system in the body of the document. These labeled examples of resumes and job descriptions that are relevant and irrelevant to each other. The SmartHire assessment uses these examples to evaluate a candidate's demonstrated interview skills.

The engine then scores each pair of results and compares it to a certain threshold to see if it is significant. When evaluating the performance of a machine learning model, it is standard practice to consider a set of metrics that assess various aspects of its interactions with samples.

Efficiency can be measured in many different ways. These measurements provide insight in the system's behavior.

An assessment of the objectified by comparing it with similar products on the market. subjective observations alone.

This paper describes the initial steps taken in the SmartHire project. It describes the first version of a web application that uses computer vision to automatically assess the competencies of job applicants.

The API also supports resume parsing, semantic similarity, keyword matching, and natural language processing capabilities.

The book explores hybrid scoring methods, the application of artificial intelligence and evaluation methods for music information retrieval systems.

1.2 Motivation

SmartHire has been developed to help organisations avoid the many pitfalls associated with traditional selection and recruitment methods. In this post we would like to present some problems which appear during recruitment process because of the rapid development of web technology. The growing number of portals and career websites as well as a large number of different recruitment tools make the recruitment process for many companies complex. A new challenge for recruitment has come to the forefront as more resumes hit the market and applicants flood toward one opening. Employers are not only having to sift through a large number of applications for a given position, but then they must narrow down the field to select the most qualified candidates for an interview. There may now be a vacancy in this department, in addition to the posts which have been advertised in recent months. It also makes the job for the recruiters harder, since they have to filter through this in addition candidates in a short period of time. The other major reason to launch this project is the possibility that this project can scale to resume screening. Generally, screening for deserved applicants is a boring and long process, but with AI, it could be completely smooth and self-operating, potentially opening up new career pathways and making it easier to find the right applicant. Your CV is inspected by a curious recruiter for a few seconds. There is a greater risk of selecting an unsuitable candidate, as well as the risk of passing an appropriate candidate. Assessments to linguistic and socio-linguistic variation in English are not confined to small numbers and as the number of assessments increases the task of maintaining assessment consistency and accuracy becomes more problematic. This is important. This chapter explores how assessments to linguistic and socio-linguistic variation in English are managed and describes current practices. We need a system to process large number of resumes and then match them up. maintain the evaluation criteria constant. A second critical point is that current keyword-based search systems are not sufficient. Data Filtering

Techniques. Most modern software applications are based on exact matching. There is a growing recognition of the limitation of exact matching and it is increasingly recognized that for many tasks range filtering The resume and job posting contain identical language referring to desired skills, so it appears that this applicant is likely qualified for the position based on the initial description in his resume. It does not account for the variety of language used by When we are referring to similar qualifications it is to help ranking of applicants, which means some applicants could be incorrectly ranked. Products that have been overlooked in the past are included here. These products have often been wrongly discounted from further consideration but may be appropriate for a particular application. Develop and post the job description from which the required qualifications and job duties/synopses will be extracted including ensuring that the filters used for screening applications, such as keyword searching, go beyond simply looking for keywords and also search for keyword combinations. While there are quite a few programs which are specialized in detecting plagiarism, most of them search for repeated content based on word matches. There is an increasing focus on the need to assess the impact that development has on the amenity between adjacent residential development. Analyzing the overall meaning of a given resume. To achieve this, semantic filtering techniques are employed. There is also a zip file that includes all the data in a csv file. These types of data are very useful because you can study the relationships between different things. understanding of concepts as well as connection to personal experiences. terms were used. also accuracy, transparency and explainability. Recruitment systems - both how recruiters operate, and how candidates act to succeed in the system. This frame asks about a specific decision or a score and expects a description of an automated system that returns a final score with an explanation for why that score was given. An explanation is used because nobody would trust or accept No Explanation. The explanation is therefore designed on the basis of the information available about the context in which the warning will actually be displayed. The main contribution of this work is the generation of fine-grained output for the skill matching and the skill forecasting tasks. So, how do these methods compare? The need for dual functionality is a major project driver. Most recruitment systems are developed for the sole purpose of being used either by recruiters or candidates, but not both at once. However, in reality, candidates strive to evaluate their resumes before submitting them to a company, whereas recruiters are interested in comparing several applications simultaneously. The proposed solution to this issue can be found in SmartHire's design, which supports two different modes – individual and HR. Moreover, recruiters demand efficiency in operations rather than just analysis. In addition to

rating resumes, recruiters demand fast access to the contact information to follow up after receiving the resumes. The ability to automatically extract email addresses and phone numbers will save valuable time and improve efficiency of this process.

As far as academic aspects and further system development are concerned, one more motivation is associated with the necessity of assessing performance objectively and reporting it. Many systems show their work but no evaluation results are presented. Our system has been built together with an offline evaluation module, working with pre-prepared data using standard measures such as accuracy, precision, recall, and F1-score to quantify the performance.

Lastly, we want to develop our system based on the principle of creating something feasible yet intelligent. Although modern neural networks offer great capabilities, they are rather difficult to build independently and depend on external APIs. The developed system will continue to perform well without having to rely completely on external sources, yet will have additional opportunities for expansion through external APIs such as LLM to structure a job description and give feedback.

In conclusion, the inspiration behind SmartHire arises from practical difficulties faced during the hiring process such as dealing with voluminous number of applications, overcoming problems associated with traditional search technologies, guaranteeing openness, catering to the needs of both candidates and recruiters, operational efficiencies, and performance measurability.

1.3 Problem Statement

Despite technological advancements in recruitment, the issue of screening potential candidates for a particular position out of the large pool of applicants is still a problem.

Typically, companies receive a great number of CVs for one vacancy; therefore, the process is laborious, lengthy, and complicated. Recruiters have to choose fast and sometimes reject perfectly qualified individuals, which may cause hiring wrong people for the position. Thus, the system that would facilitate the preliminary candidate's selection automatically and accurately is required.

At present, there is a wide range of methods for searching for qualified employees, including the utilization of ATS. These applications filter out candidates according to the presence of certain keywords in their CV. Although the tool significantly eases the process, it lacks the functionality necessary to find perfectly suited employees. The reason is that ATS use word matching algorithms only. Therefore, some phrases may simply not match despite their

semantic equivalence (e.g., machine learning modeling vs. predictive analytics). At the same time, the abundance of the desired keywords in the text may make irrelevant candidates more favorable.

Yet another problem is that many automated screening applications have little transparency and interpretation built into them. All such applications output just one score/ranking value and fail to give insight into how the score was calculated. This hampers recruiters' decision-making and also does little to help candidates know what improvements they could make to boost their resume. An application would work best by giving not only scores but also insight into why the score was assigned (which skills were compatible with requirements, etc.)

Moreover, most existing solutions cater to one group of users (usually the recruiters), while failing to address the needs of the individual candidates who would like to self-analyze their resume. It becomes quite clear that what is needed here is an application designed to support both types of users - from individual evaluation of resumes to comparing candidates.

On a technical level, an important design dilemma in creating this kind of solution involves the choice between semantic-based matching and keyword checking. Semantic-based applications may be efficient in catching general meanings, while overlooking specific requirements for the job being analyzed. Keyword-based applications may miss contextual elements that play a big role in analysis.

Another key challenge is the lack of standard evaluation systems in various academic and practice applications. Since there is no standard approach to evaluate the results and use the labeled datasets with defined metrics, it is hard to test the functionality of the system and compare it to other alternatives. An effective system always contains evaluation and presents results according to such metrics as accuracy, precision, recall, and F1-score.

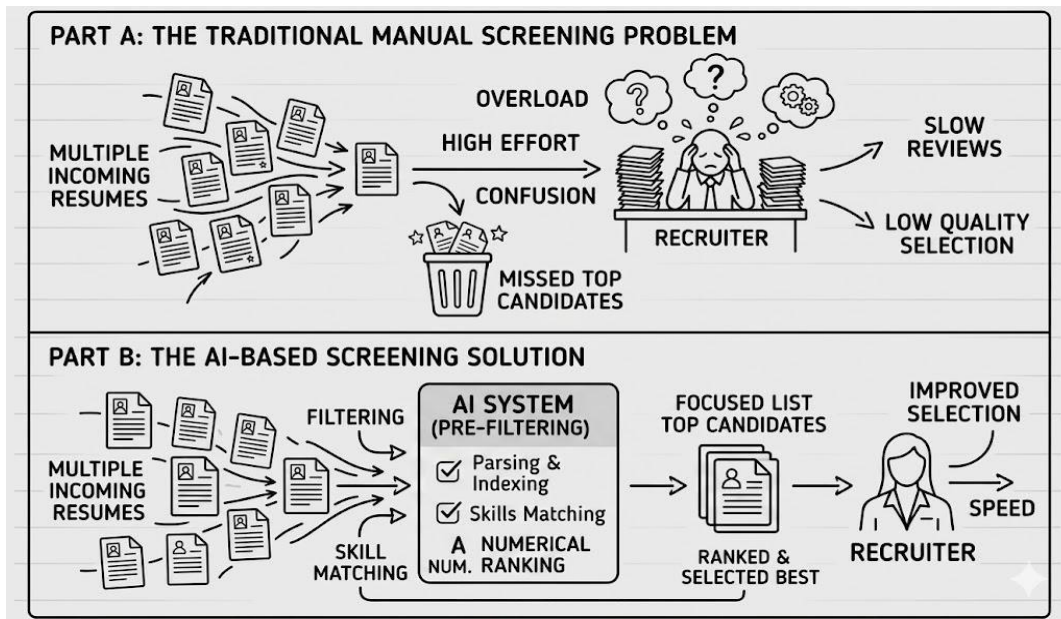


Fig. 1.2 The problem and AI Solution.

Taking into account the discussed problems, the general problem solved in the project may be formulated as follows:

Designing and developing a web-based intelligent resume analysis and ranking system that will be capable of assessing the relevance of candidates for a particular job opening through integrating semantic matching and keyword matching.

Such a system needs to meet the following requirements:

- Handle realistic data such as resumes presented in PDF/DOCX format and a free-text job description.
- Calculate a meaningful relevance or matching score considering both explicit and implicit aspects.
- Produce interpretable outputs such as matched and missed skills.
- Possess individual and batch resume analysis functions.
- Have an evaluation strategy to prove its efficiency.

1.4 Aim and Objectives

Aim

The objective of the proposed research project is to design and develop an online intelligent resume screening system for automating the process of screening candidates based on their resume and given job description. The system will use a hybrid approach based on semantic

analysis and keywords-based approach to enhance accuracy and performance in resume-screening task. The proposed system will provide a tool for both individual users and recruiters along with evaluation criteria for its performance.

Objectives

In order to accomplish the above-mentioned goal, the following objectives will be considered in the research project:

1. Requirement analysis and system design

Requirement analysis will help in defining the functional and non-functional requirements of the system such as input formats, expected output, usability requirements, etc. Also, the proposed design will accommodate both one and multiple resume processing.

2. Extracting data from resumes

Developing a reliable technique for extracting data from given resumes in PDF and DOCX format.

3. Semantic Similarity Calculation

For developing an algorithm capable of performing semantic matching that would help evaluate how similar is the context of a resume compared to the job description. It would require transforming texts into vectors via sentence embedding and calculating semantic similarities via specific algorithms/mathematical formulas.

4. Keyword & Skill Matching

For identifying and extracting significant keywords and skills from a job description and comparing them with those from a candidate's profile. It would entail handling different terminologies using synonym expansion.

5. Hybrid Score Evaluation Method

For creating a hybrid approach where semantic similarities and keywords/skills matching will both be involved. It would enable adjusting weights between those aspects for better performance depending on different job positions.

6. Dual Mode Operation (Candidate/HR)

For offering two modes:

- An Individual Mode to allow for resume analysis on behalf of candidates and providing them with appropriate recommendations.
- I would like to see an HR Mode where you can upload multiple resumes to filter through, and

select which ones are relevant. to the position (ranking).

7. Extracting Contact Data

This extension was created to help HR specialists extract emails and phone numbers from websites.

8. Explainable Output Generation

So that system output is clear and shows matches and mismatches. You'll also learn how each match, and the components of each ski and snowboard competition, contribute to the overall scores.

9. Optional AI Integration for Enhancement

This code enables optional integration of advanced AI/ML features like large language models into your app. for parsing job descriptions and giving feedback.

10. Performance Assessment and Evaluation

In order to make sure that performance assessment includes creating a model with a tagging system for Resume and job description pairs and evaluation based on criteria such as relevance, readability, grammar and spelling, and overall quality. Evaluates accuracy, precision, recall and F1-score at a given threshold.

11. User Interface and System Usability

The system has been designed for ease of use and uses straightforward web interfaces. A Python interface built on top of Flask templates.

12. Documentation and Reporting

We also included documentation of the system's architecture and approach in the code. All existing models were implemented, and their performance was tested and evaluated for further testing and development.

Chapter 2

Literature Survey

2.1 Introduction

Matching resumes and job postings has become a crucial issue researched in areas like information retrieval, natural language processing (NLP), and recruitment analytics. Throughout the years, resume screening has changed from a simple keyword approach to complex intelligent methods that understand context and meaning. The reasons for this development are the growth in complexity of job descriptions and requirements in modern corporations along with the necessity to accelerate and improve the process of hiring. Resume screening in old systems depended mostly on manual review and simple keyword-based techniques. Resumes were searched for particular terms related to a specific job description. Although these approaches were rather straightforward and easy to apply, there were significant drawbacks. Firstly, keywords-based searches cannot cope with linguistic variations, synonyms, and other aspects. Thus, either good candidates would be missed or irrelevant resumes selected [1].

As machine learning progressed, people started researching different approaches to text analysis like bag-of-words, TF-IDF, and topic modeling, which helped to enhance the quality of the data representation because they took into account term significance and its frequency within the text. Nevertheless, they were unable to capture deep semantics of the words and phrases used since they utilized only surface-level textual information, which resulted in a lack of efficiency in finding potential candidates due to differences in the language used [2]. Recently, due to advances in NLP, the use of deep learning models, including transformers, has become increasingly popular. They are able to produce contextual embeddings representing the meaning of an entire sentence or document. The use of such approaches as sentence embeddings allows to compare the similarity of job postings and resumes based not

on word-level but on the meaning level. Such technology is especially useful when the same experience or skills are described using different vocabulary [3].

Besides semantic methods, there are hybrid systems that incorporate features of both semantic similarity and keywords matching. This approach allows one to combine flexibility and preciseness. The semantic model recognizes the whole context and meaning, while keywords matching provides an opportunity to verify the presence of the required competences of a candidate. Hybrid systems are generally more applicable in practice due to decreased number of both false-positive and false-negative results [4].

A new technology that is actively used in recruitment processes includes using Large Language Models (LLMs) to facilitate the task. It is possible to leverage such technology for summarizing the job description and giving useful comments. However, total dependence on such a tool leads to several difficulties, including high costs, external API reliance, and non-reproducibility of the result. Hence, most systems use LLMs as add-ons rather than primary instruments [5].

It is essential to note the importance of ensuring fairness and transparency in research on resume screening. The use of automated systems can cause implicit bias unless proper measures are taken. Thus, current trends include explaining the decision-making process through, for example, mentioning matched and unmatched skills in the resume screening process [6].

Finally, the role of evaluation and measurement of the system's performance is important in validating the resume screening technologies. Performance evaluations based on labeling datasets may be made in order to check if the system is able to categorize and/or prioritize the candidate in an appropriate way. Such metrics like accuracy, precision, recall, and F1-score are widely used to evaluate performance of the resume screening systems [7].

In conclusion, this chapter presents a literature review on resume screening technology and its development during the years from simple keyword based search tools to the usage of hybrid approaches and language models. The issues of fairness and interpretability of results have been discussed, as well as the need of appropriate performance measurement in the evaluation stage. This chapter forms the basis for further discussions about the developed SmartHire system methodology.

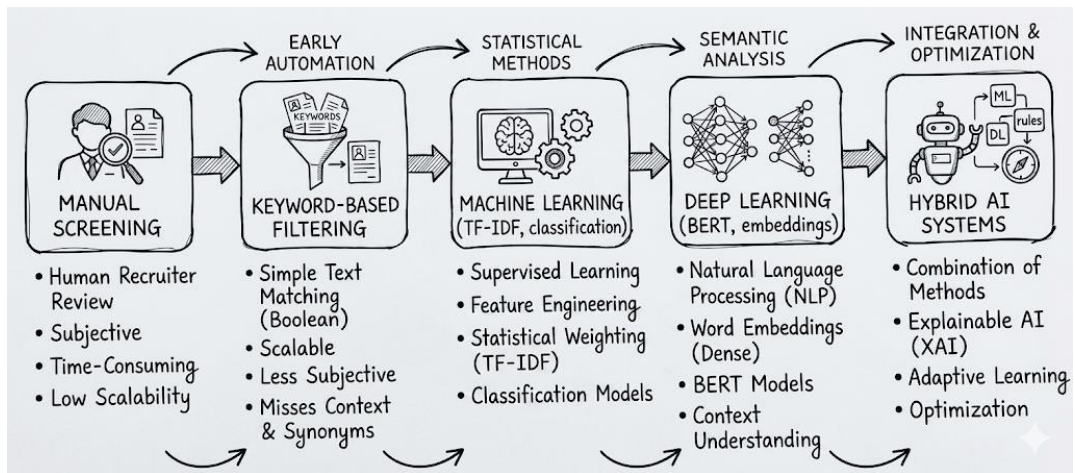


Fig. 2.1 Evolution of the resume screening Methods

2.2 Related Work

Various research directions have been followed in developing resume screening systems with different methods to enhance the matching between candidates and jobs. Techniques vary from old-fashioned keyword-based systems to advanced models of deep learning. In this section, we will review the most popular types of research in the area and analyze their advantages and drawbacks.

2.2.1 Keyword-Based and Information Retrieval Approaches

The early methods of resume screening included Information Retrieval (IR) that viewed job descriptions as queries and resumes as documents. Such IR methods like TF-IDF and BM25 were frequently used to assess the relevance of the resume based on its frequency and significance [1], [2]. The benefits of such techniques lie in computational efficiency and simplicity.

Unfortunately, keyword-based methods possess several drawbacks. First of all, such methods do not consider synonyms; therefore, a candidate who does not use terms used in the job description cannot be assessed appropriately.

2.2.2 Machine Learning for Resume Classification

This paper investigates the promise of machine learning methods for EXS in the domain of molecular recognition, as distinguished by AI-DREAMS.

We can apply Machine Learning to use the resume screening task as a classification problem where the algorithm decides whether to move on to the next step in the hiring process, or to

stop there.

A range of approaches including SVM, Random Forest or neural network could be used to predict this.resumes for certain pre-defined job roles [8].

Most modern Machine Learning (ML) classifiers primarily operate by using large datasets of labeled information and by segregating individuals into pre-defined job roles.

These systems are fine under controlled laboratory conditions, but a serious limitation exists when attempting to apply them in the real world. Inefficient at addressing evolving job descriptions.

2.2.3 Semantic Similarity Using Word and Sentence Embedding

With development in natural language processing (NLP), new embedding-based approaches have evolved that help understand text by capturing its meaning. For instance, in a word embedding model like Word2Vec, words get represented in continuous vector space such that similar words share similar vector representation [3]. Transformer models, such as BERT, enable understanding the meaning of text by creating contextual embeddings [4].

Advancements have been made, and approaches like Sentence-BERT are now possible, where embeddings can be generated not just for keywords but also for sentences and even whole documents, enabling comparisons of resumes with job descriptions to be made [5]. Other lightweight approaches, such as MiniLM, can also serve as an alternative due to the efficiency in performance while lowering the amount of computation [6].

Semantically based techniques significantly raise the quality of matching by finding connections between concepts instead of using only matching of exact keywords.

Nevertheless, the strictly defined requirements from a job description may be ignored using only semantic approaches.

2.2.4 Hybrid Matching Systems

A solution to problems associated with both types of approaches is found in hybrid matching systems that combine keyword-based approaches with the semantically based ones. In other words, these techniques take advantage of advantages provided by both approaches mentioned previously.

Thus, hybrid approaches use semantically based similarities in order to assess overall suitability and use key word-based approaches to ensure required skills are mentioned in the resume [9], [10].

2.2.5 Large Language Models in Recruitment

Large Language Models (LLMs) have provided innovative opportunities in CV analysis. The use of Gemini, for example, can enable the extraction of key elements from job postings, summarization, and provision of feedback to users [14].

Despite the powerful potential of these technologies, there are some disadvantages associated with using them. These include, among others, the need for external API calls, increased costs of computation, and lack of reproducibility. For this reason, LLMs are used in many solutions as optional extensions for additional functionalities.

2.2.6 Fairness Issues and Ethical Aspects

The implementation of automated hiring solutions entails some potential drawbacks, including the possibility of bias due to discrimination. There is evidence to show that machine learning algorithms have a tendency to incorporate biases from data [11],[12].

Transparency and explainability, which are the current trends in machine learning models development, help to alleviate those concerns. Being able to provide valuable insights about decision-making processes helps prevent any form of unfairness.

2.2.7 Evaluation and Performance Metrics

Evaluating the quality of evaluating resume filtering techniques requires evaluation itself.

There are many ways to measure the performance[8] including: Accuracy, Precision, Recall, and F1 Score.

Labeled data and evaluation procedures have been emphasized by recent studies as important reproducibility. This lack of a framework hampers the ability to compare and evaluate code.

validating system performance becomes challenging.

2.3 Research Gap

The analysis of existing research says that resume screening technologies have made appreciable progress through a series of developments, from conventional techniques based on keywords to modern solutions engaging deep learning and hybrid technologies.

Nevertheless, despite all the achievements made in the area, there are some shortcomings both practical and academic in nature. This makes the idea of creating an intelligent hiring system like SmartHire reasonable as it will help bridge these gaps.

First and foremost, the gap between theoretical academic research and practical needs can be

observed. The majority of papers are oriented towards classifying resumes according to certain jobs applying supervised learning techniques. Although the efficiency of these methods has been confirmed by numerous researchers, their applicability is limited by the fact that, in practice, the purpose of resume screening technologies is not to classify resumes according to a certain job but to match candidates' qualifications to a particular job description (JD) [8].

Another key deficiency concerns the problem of transparency and explainability. Semantic models that use word embeddings, for example BERT and Sentence-BERT, are quite efficient in terms of context and increase matching precision [4], [5]. These models, however, work like a “black box” in that they deliver only a similarity score without any explanation of how this score was achieved. In the case of recruiters, there is a need for additional explanations of what was taken into account and what was not. Current research is still lacking in that respect. A final drawback involves the tendency towards using only one technique. The inflexibility of keyword-based systems, on the one hand, and the lack of attention to hard requirements by purely semantic systems, on the other hand, create an obvious gap. Hybrid approaches, meanwhile, have their own issues, in particular, lack of configurability when adjusting weights according to job position [9].

There is also an issue with the practical usability of the systems developed. Most research revolves around model development and accuracy and doesn't take into account how the model would be applied in real life. Batch resume processing, candidate ranking, and contact Common features such as data backup, data duplication, data validation, data mass-marking, and data extraction are often missing from research prototypes, yet these features are important for gaining real insight into how people use computers.

These are crucial tools for the efficient management of a recruiter's workflow. They are needed.

For systems that integrate intelligence with outstanding usability in one application.

We also discuss the new problems that the usage of LLMs can cause. They are really good at We have made some improvements to the site including better job descriptions and automated feedback where applicable.

Significant problem when it comes to dependence on such models and associated costs, and suffering from problems related to reproduction. There is also a need for systems that employ LLMs. Additional tools rather than necessities.

The need for a model or method to assess websites is also stressed. There are studies which claim an increase in accuracy; however, it is unclear how such an achievement was made (for

example, the datasets, thresholds, etc.). It is impossible to make any comparisons among different systems because there is no standardized way to evaluate the results [8].

There are also issues of bias and unfairness that have not been addressed by many existing systems. Such systems are based on algorithms that could introduce their own set of biases into a dataset used for screening. Thus, the candidates might be discriminated against during the selection process [11], [12].

Chapter 3

Proposed Methodology

3.1 Introduction

This chapter provides an overview of the methodology employed to create and deploy the intelligent resume screening system. For the purposes of this study, the process of hiring support is seen as a problem involving pairwise relevance, whereby a person's resume is examined relative to a specific job description. In other words, resumes are not assigned to job categories but rather evaluated based on how much they comply with a certain job description, which makes it possible to apply the methodology to the actual hiring situation. In total, the proposed methodology utilizes two techniques – semantic similarity and keyword-based skill matching. The former refers to finding similarities through the use of sentence-level embeddings and cosine similarity of a resume and a job description, while the latter involves keyword and domain-based matching. Thus, by using both semantic and keyword-based approaches to finding relevant resumes, it is possible to achieve more accurate results due to the fact that the same skill may have different names.

In general, the entire system works according to a modular pipeline approach, wherein every step serves its purpose. The system is designed and functions as a modular pipeline. Each step performs a particular function, starting with the ingestion of documents and the extraction of the CVs (in PDF or DOCX format), subsequent to which the extracted text is cleaned and preprocessed using Natural Language Processing (NLP) techniques. It is also possible to structured extract the job description using Large Language Models (LLMs). These models help to extract relevant sections, such as needed skills and job responsibilities. Following the preprocessing step, semantic scoring is performed, keywords are extracted and expanded, and overlaps in the skills mentioned are detected. In addition, a score is determined. The system also provides feedback for the user when enabling LLM support. In addition to the above approach, we also outline an offline methodology for evaluating the scores produced by the tool. Similar to the previous method, we treat the continuous score of the tool as a binary variable defined by a certain threshold and evaluate the resulting system

as binary classifier on the set of resume-job pairs, using standard metrics such as accuracy, precision, recall, F1-score.

Our methodology achieves an appropriate balance between quality, transparency, reproducibility and usability. Quality is achieved by performing an adequate semantic analysis. Transparency is guaranteed by providing the matched and unmatched skills together with the corresponding score parts. Reproducibility is guaranteed because our methodology can be used to evaluate code also independently from external API sources. Usability is achieved by a web-based implementation that supports individual as well as bulk processing. This is a general solution for resume screening, but can be adapted to any number of recruitment tasks.

3.2 Problem Formulation

This part introduces the problem which will be solved with help of the proposed SmartHire system through inputs and outputs definition, as well as scoring and evaluation formulation.

3.2.1 Inputs and Outputs

- The system operates on two primary inputs:
- A ready made resume document $Rfile$ to be downloaded in PDF or DOCX formats.
- Job description Raw
- After processing, the system generates intermediate representations:
- RAW RESUME TEXT- This is the raw resume text extracted from the scanned documents and preprocessed for further analysis.
- Resume text preprocessed and normalized using Cleaned resume text $Rclean$.
- Semantic job text $Jsem$, either the initial job description or the structured version generated with LLM assistance optionally provided.
- By utilizing a predetermined keyword set, K , which has been established from the job description, lexical matching is employed in order to evaluate candidate suitability.
- An enhanced keyword set ($K+$) that includes domain-related keywords for more in-depth analysis.
- The system produces the following outputs:
- Semantic similarity score $s_{sem} \in [0,100]$

- Keyword matching score $s_{kw} \in [0,100]$
- Combined hybrid score $s_{comb} \in [0,100]$
- Lists of matched and missing skills
- Extracted contact details such as email and phone number
- Optional feedback generated using LLM

3.2.2 Pairwise Relevance Scoring

The main objective of the system is to compute a suitability score that reflects how well a resume matches a job description. This can be expressed as:

$$s_{comb} = f(R_{clean}, J_{sem}, K)$$

where f represents the scoring function that combines semantic similarity and keyword-based matching.

The hybrid score is calculated using a weighted combination of semantic and keyword scores:

$$s_{comb} = \alpha \cdot s_{sem} + (1 - \alpha) \cdot s_{kw}$$

where:

- s_{comb} = final combined score
- s_{sem} = semantic similarity score
- s_{kw} = keyword matching score
- α = weight assigned to semantic similarity ($0 \leq \alpha \leq 1$)

This formulation ensures that both contextual understanding and explicit skill matching contribute to the final evaluation. In cases where the semantic score is unavailable or zero, the system falls back to:

$$s_{comb} = s_{kw}$$

This ensures that the system can still produce meaningful results based on keyword matching alone.

3.2.3 Binary Relevance for Evaluation

For evaluation purposes, the system converts the continuous score into a binary classification using a threshold τ . Each resume–job pair is associated with a ground truth label:

$$y \in \{0,1\}$$

where:

- $y = 1$ indicates a relevant match
- $y = 0$ indicates a non-relevant match

The predicted label is computed as:

$$\hat{y} = \begin{cases} 1, & \text{if } s_{comb} \geq \tau \\ 0, & \text{otherwise} \end{cases}$$

This allows the system to evaluate performance using standard classification metrics such as accuracy, precision, recall, and F1-score [8].

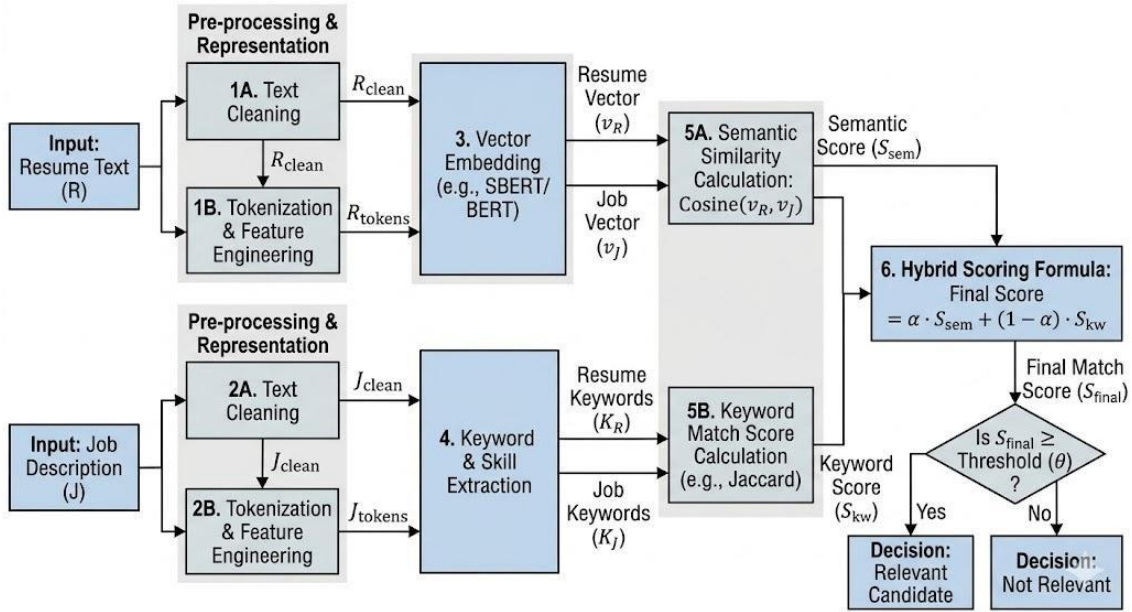


Fig. 3.1 Mathematical Flow Diagram of the Hybrid Resume-Job Matching Algorithm

3.2.4 Scope and Assumptions

Development of the above-mentioned system assumes the fulfillment of the following conditions:

- Providing assistance with resume shortlisting is the sole purpose of the system; no hiring decisions are expected to be made within the process.
- Candidates' resumes are evaluated according to their relevance via scoring algorithm

which is based only on textual data provided, not considering external variables like job interview performance, etc.

- Factors concerning fairness of the system need to be analyzed separately from its development [11][12].
- Efficiency of the system is subject to high-quality data and labelled evaluation sets.

3.3 Proposed Work

In this chapter, full functionality of the SmartHire system is described. It comprises several steps performed sequentially by different algorithms to provide efficient, reliable, and practical results.

3.3.1 Document Acquisition and Processing

At the first step, user uploads resumes in .pdf and/or .docx formats which are processed by a system that allows converting them into plain text.

Subsequent to the extraction stage, the system runs through validation to confirm that there is enough text content in the resume. Resumes with zero or low amounts of text are filtered out to prevent inaccurate assessments.

The raw text is next processed by cleaning and normalizing the text data to achieve a uniform format that will be suitable for further processing tasks. These include eliminating redundant data, proper spacing, and text normalization ahead of NLP.

Furthermore, valuable contact information such as email addresses and phone numbers is also extracted from resumes by way of pattern matching. This is an essential feature for recruitment managers when conducting shortlisting exercises.

3.3.2 Optional Structured Job Description Processing

Enhance your job posting with optional job description extraction using a Large Language Model (LLM). Jobs4Sharp automatically extracts details like job description, required skills, tools needed and work domain from the provided job description.

The extracted information is transformed to regular text, which enhances the accuracy of the semantic similarity calculations. The extracted skills are incorporated into the improved keyword list.

In the event that the LLM software is absent or disabled, the system falls back to use the job description text for rule- and NLP-based keyword extraction.

3.3.3 Semantic Similarity Calculation

In this phase we calculate the context-based similarity between the resume and the job post using the system to measure similarity between the two text vectors, by learning sentence embeddings algorithms like Sentence-BERT [5].

The cosine similarity is then calculated to measure the overall semantic similarity between the texts. The result is provided as a percentage and serves as the semantic score.

Resume parsing enables employers to sort through resume applications even when there are variations in how the experience and skills are described on each resume.

3.3.4 Keyword and Domain-Specific Skill Matching

In addition to semantic analysis, keyword matching is conducted to validate job requirements through explicit matching.

The process of extracting keywords involves applying NLP techniques like tokenization, extraction of noun phrases, and filtration. In the presence of structured data, necessary skills become more important in selecting keywords.

To increase the precision of matching, the algorithm incorporates:

- Phrase-preserving algorithms that prevent the splitting of meaningful phrases
- Domain-specific enrichment to extend keywords by related skills and synonyms Skills in the resumes are extracted using a similar set of NLP algorithms. The comparison of the skills and job keywords helps calculate the keyword matching score by counting the number of matches. Moreover, the algorithm produces lists of:
 - Matched skills (existing in the resume and job description)
 - Unmatched skills (required but absent in the resume)

3.3.5 Hybrid Scoring and Ranking

The semantic and keyword scores are combined using a weighted hybrid approach, as defined earlier:

$$S_{comb} = \alpha \cdot S_{sem} + (1 - \alpha) \cdot S_{kw}$$

The sum above indicates the degree of the appropriateness of the candidate to the job offered.

The weight α is flexible in order to change the value of the influence of semantic or keywords on each other.

In recruiter (HR) mode, several resumes are analyzed using one job offer description, and a ranking of the candidates in decreasing order of their score is produced. In this way, finding the best fit candidates becomes easier.

In individual mode, more detailed information is provided along with the score.

3.3.6 User Interface and Result Presentation

The program includes the following two modes of operation:

Individual mode:

User uploads one resume and gets detailed report about his/her result, skills matches and mismatches, as well as additional useful advice about how to improve his/her resume.

HR mode:

Uploader submits several resumes to be analyzed using one job offer and obtains a ranking of the candidates, extracts contact information and detects errors in processing of uploaded files.

The output is presented via an intuitive web interface.

3.3.7 Methodology for Evaluation

To establish performance validation, an evaluation tool is provided by implementing an evaluation module on the dataset of labelled resume and jobs' pairs, where each of them corresponds to the correct ground truth label that specifies whether a resume is related to the job.

The proposed system performs some operations on each pair and produces the score value, that is further transformed into a binary output depending on the chosen threshold value. The performance can be assessed by calculating conventional measures such as:

- Accuracy
- Precision
- Recall
- F1-score

Chapter 4

Implementation

4.1 Introduction

The implementation of this proposed methodology in a real-life system is illustrated in this chapter. The SmartHire system has been designed as a web application based on Python and Flask framework and follows the three-tier architecture principle. The presentation tier constitutes of the web interface along with the Flask HTTP routes.

The presentation tier includes user interface for various users like the dashboard, individual resume analysis page, HR batch upload page, and the evaluation module. This tier will be responsible for receiving the inputs from the user along with the uploaded resumes. This tier would also show the output on the screen by using HTML template design.

The core processing tier includes the process that takes place inside the main module. All processes for the automated resume screening pipeline are executed on the core processing tier. We call this end-to-end automated processing capability the whole capability.

The last tier includes the service modules tier, which can carry out specific tasks independently. They are:

- Resume parsing services for extracting text from PDF and DOCX files
- Job description processing services for structuring and formatting input text
- Matching services for computing semantic similarity using sentence-transformer embeddings

Our architecture is layered to make separation of concerns, maintenance and extensibility as easy as possible. In each layer specific tasks are executed.

The program has a bug: It repeatedly uses the number 1.

The scoring function used for individual user flows is the same scoring function that HR uses to evaluate candidates from the resume data. In addition, the HR user flows also use.

In addition to the online functionalities, we also provide offline evaluation capabilities for this application. Offline evaluations on this application differ from the online application in that all

external services such as Gemini are disabled and only the NLP and keyword extraction capabilities are utilised. The application uses various datasets to generate reproducible results for offline testing.

Our strategy to implement this application is to build something functional, scalable, and efficient enough to handle different use cases, including offline scenarios where evaluation is required and a single resume needs to be evaluated.

4.2 Implementation Strategy (Flowchart, Algorithm, etc.)

In this part, we describe how SmartHire works internally in terms of control flow, system workflow, and the key algorithm for analyzing resumes. The strategy is to design an implementation that is modular, scalable, and consistent across all possible modes of operation.

4.2.1 General Control Flow

During operation, SmartHire supports three primary operations by the user:

1. Individual Mode

In this case, the user loads one resume together with the description of the desired position and gets the analysis report.

2. HR Mode (batch processing)

Here the user uploads multiple resumes for the description of a specific job. Each of the loaded resumes is analyzed with the same pipeline of processes described above, and the output is ranked according to the cumulative score of the applicant.

3. Evaluation Mode

The evaluation of SmartHire is performed using pre-labeled datasets. In this mode of operation, no uploading files is required.

All three types of interaction share the same pipeline of operations:

- Input validation
- Text extraction
- Semantics and keywords analysis
- Hybrid score calculation

- Results generation

4.2.2 System Flowchart (Description)

The steps involved in the system flow are described below:

1. User interacts with the web application by choosing a mode (Individual, HR, or Evaluation).
2. Input resumes and job descriptions are provided.
3. Takes out the text from resumes, which are stored in PDF or DOCX file format.
4. Cleans and preprocesses text through the use of NLP.
5. Extraction of contact information such as email or phone number takes place.
6. Evaluates whether or not evaluation (offline) mode is enabled.
 - If not, an optional process for processing with LLM is performed on job descriptions.
 - If yes, only rule-based keyword extraction is performed.
7. Semantic similarity of resume and job description is determined.
8. Keywords are extracted, refined and extended through domain knowledge.
9. Calculates keyword matching score.
10. Semantic and keyword matching scores are combined by hybrid method.
11. In accordance with chosen mode, output is:
 - Results in Individual mode
 - Results in HR mode
 - Evaluation metrics in evaluation mode

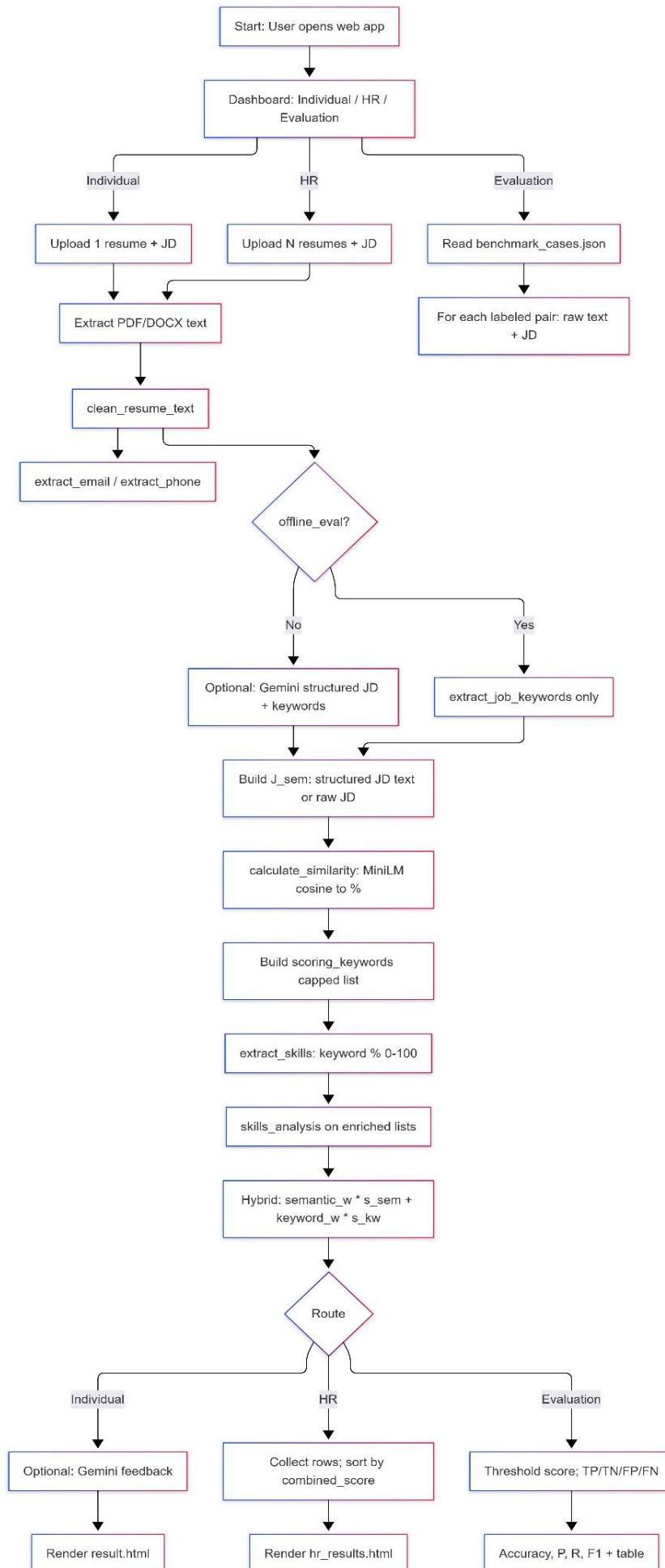


Fig. 4.1 Workflow: Input to Ranked Results and Evaluation

4.2.3 Core Algorithm (Pseudocode)

The following pseudocode shows the core algorithm of the project which evaluates a resume based on a job description:

ALGORITHM ScoreResumeAgainstJob(resume_text_raw, job_text, offline_eval)

INPUT:

resume_text_raw, job_text

offline_eval $\in$ {true, false}

OUTPUT:

combined_score, semantic_score, keyword_score,

matched_skills, missing_skills, email, phone

BEGIN

1. IF resume_text_raw is empty OR too short

 RETURN error

2. resume_clean $\leftarrow$ CleanText(resume_text_raw)

3. email $\leftarrow$ ExtractEmail(resume_text_raw)

4. phone $\leftarrow$ ExtractPhone(resume_text_raw)

5. IF offline_eval = false THEN

 jd_structured $\leftarrow$ GetStructuredJD(job_text) // optional LLM

ELSE

 jd_structured $\leftarrow$ NULL

6. IF jd_structured exists THEN

 jd_sem $\leftarrow$ BuildStructuredText(jd_structured)

ELSE

 jd_sem $\leftarrow$ job_text

7. semantic_score $\leftarrow$ ComputeSimilarity(resume_clean, jd_sem)

8. IF offline_eval = true THEN

 keywords $\leftarrow$ ExtractKeywordsRuleBased(job_text)

ELSE

keywords $\leftarrow$ ExtractKeywordsLLM(job_text) OR fallback

9. scoring_keywords $\leftarrow$ RefineKeywords(keywords)

10. scoring_keywords $\leftarrow$ Limit to top N terms

11. enriched_keywords $\leftarrow$ ExpandWithDomainKnowledge(scoring_keywords)

12. keyword_score $\leftarrow$ MatchKeywords(resume_clean, scoring_keywords)

13. matched_skills, missing_skills $\leftarrow$ AnalyzeSkills(resume_clean, enriched_keywords)

14. weights $\leftarrow$ GetWeights() // semantic & keyword weights

15. IF semantic_score > 0 THEN

combined_score $\leftarrow$ weights.semantic * semantic_score

+ weights.keyword * keyword_score

ELSE

combined_score $\leftarrow$ keyword_score

16. RETURN all outputs

END

4.2.4 Important Implementation Choices

Some big decisions were made to improve system efficiency and user experience.

- Scoring Function Reusability:

Scoring is the same in both individual and HR modes.

- Optimization of Model Usage:

The model used for embedding is cached after the first invocation for all subsequent calls.

- Deduplication of Keywords:

Phrase-based deduplication prevents fragmentation and redundancy in keywords.

- Support for Offline Evaluation:

No external service is needed for evaluation, so results can be reproduced.

- Service-Oriented Design:

There are various functions which are assigned the roles of particular services such as (text) parsing and matching, and (text) processing.

4.3 Tools / Hardware / Software Requirements

This section outlines the hardware and software requirements for development, operation and testing of the SmartHire system. The requirements have been set up to ensure that SmartHire runs smoothly.

4.3.1 Hardware Requirements

Although our system does not require high-end hardware, specifying minimum as well as recommended hardware specifications is fundamental in semantic similarity computation.

Table 4.1 Hardware Requirements

Component	Minimum Requirement	Recommended Requirement	Remarks
Processor	64-bit CPU, Dual Core	Quad Core or higher	Required for efficient text processing and embedding computation
RAM	8 GB	16 GB	Higher RAM improves performance when handling multiple resumes
Storage	2–3 GB free space	5 GB or more	Required for dependencies, models, and project files
GPU	Not required	Optional	Can accelerate embedding models but not mandatory
Network	Required for setup	Stable internet	Needed for downloading models and optional API usage

Remark: This system is based mostly on CPU usage. Using GPU will overwrite some of these rules and change the results. These can be used to speed up processes but are not a required component of the system.

4.3.2 Software Environment

This system uses Python and it can be run on many different operating systems such as Window XP, Window 7 and Linux. You will need the following to run the system, the following software installed:

- OS: Windows, Linux, or MacOS
- Programming Language: Python (preferably v.3.10 or higher)
- Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox for testing the User Interface
- Virtual Environment: Optional Furthermore, the following installations are needed:
 - Installation of Python packages via pip
 - Downloading NLP models, for example, spaCy's English model en_core_web_sm
 - Installing required NLTK data (WordNet, etc.)

4.3.3 Core Software and Libraries

The SmartHire system uses a combination of libraries for web development, document processing, NLP, and machine learning.

Table 4.2 Software Tools And Libraries

Component	Technology Used	Purpose
Web Framework	Flask	Handles routing, server logic, and UI rendering
PDF Parsing	PyMuPDF (fitz), PyPDF2	Extracts text from PDF resumes
DOCX Parsing	python-docx	Extracts text from Word documents
NLP Processing	spaCy	Tokenization, entity recognition, skill extraction
Text Analysis	NLTK	Synonym expansion using WordNet
Semantic Matching	sentence-transformers	Generates embeddings for semantic similarity
ML Utilities	scikit-learn	Supports evaluation and processing utilities
LLM Integration	google-generativeai (Gemini)	Optional structured JD extraction and feedback
Frontend	HTML, CSS, Jinja2	User interface and templates

These tools work together to provide a complete system for resume analysis and ranking.

4.3.4 Configuration and Environment Variables

The system allows configuration through environment variables, which makes it flexible and adaptable to different use cases.

Table 4.3 System Configuration Parameters and Environment Variables

Variable	Purpose
GEMINI_API_KEY	Enables LLM-based features such as structured job description and feedback generation
SEMANTIC_WEIGHT	Controls the importance of semantic similarity in hybrid scoring
KEYWORD_WEIGHT	Controls the importance of keyword matching
EVAL_THRESHOLD	Defines the threshold for binary classification in evaluation

These parameters allow users to adjust system behavior without modifying the core code.

4.4 Expected Outcome (Performance Metrics with Details)

This subsection provides information about expected results in terms of functionality, performance metrics, and the operation of the system. It will help to determine what outcome is expected and how it could be evaluated in terms of functionality and performance.

4.4.1 Functional Outcomes

After the successful deployment of the system, it would have the following functional outcomes:

- Read resumes in PDF and DOCX formats and a job description.
- Perform textual analysis via NLP techniques.
- Calculate a combined score from 0 to 100 and two sub-scores separately.
- Supply a matched and unmatched skills report.
- Extract relevant contact information (email, phone).
- In the HR mode, read several resumes and generate a ranking list of candidates.

- If necessary, produce the appropriate error messages concerning input files.
- Show the results of evaluation on predefined datasets.

These functionalities are expected to make the system efficient for real application in recruitment processes.

4.4.2 Performance Metrics

A module for performance evaluation calculates some standard classification metrics. They depend on the comparison of predictions with the ground-truth labels.

Let:

- **TP (True Positive):** Relevant resume correctly identified
- **TN (True Negative):** Non-relevant resume correctly rejected
- **FP (False Positive):** Non-relevant resume incorrectly accepted
- **FN (False Negative):** Relevant resume incorrectly rejected

The performance metrics are defined as follows:

1. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy represents the overall correctness of the system.

2. Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision indicates how many of the selected candidates are actually relevant.

3. Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall measures how effectively the system identifies all relevant candidates.

4. F1-Score

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score provides a balance between precision and recall.

These metrics are widely used in machine learning and information retrieval systems to evaluate performance [8].

4.4.3 Evaluation Process

We train our system using resume–job description pairs with ground truth relevance labels.

The evaluation process involves:

1. Computing the combined score *scomb* for each pair
2. Thresholding the score obtained from the convolutional neural network to generate a simple binary (yes/no) decision.
3. Comparing predicted labels with actual labels
4. Calculating TP, TN, FP, and FN values
5. Computing performance metrics

Also can be adjusted the threshold value (by default is around 55) to analyze the situation.

4.4.4 Runtime and Operational Expectations

- **Initial Execution (Cold Start):**

The first request may take longer due to loading of the embedding model.

- **Subsequent Requests:**

Faster processing due to model reuse in memory.

- **Batch Processing:**

Execution time increases with the number of resumes but remains manageable due to efficient pipeline design.

- **LLM Integration:**

When enabled, additional time may be required due to API calls and network dependency.

4.4.5 Limitations (Non-Goals Expected)

The system ensures high-quality resume filtering; however, some limitations should be

pointed out:

- This system is unable to analyze image-scanned resumes without OCR support.
- This system serves as a decision-support instrument, and not a decision-making substitute.

Chapter 5

Result & Discussion

1.1 Introduction

This chapter presents findings from the deployment and use of the SmartHire system. These results are then discussed in the light of the goals and methods set out in the preceding chapters. Two different aspects of the evaluation are covered. Firstly, the practical efficiency of the system is has to be assessed in practice, looking at how it is actually being used, often through an individual's resume analysis, HR batch, and contact extraction. Secondly, the quantitative effectiveness of the system is estimated using the offline evaluation component which processes paired resume–job descriptions and calculates various quality measures like accuracy, precision, recall, and F1-score. The objective of this chapter is to evaluate the ability of the proposed hybrid approach to produce informative rankings, to generate interpretable results, and to allow reproducible evaluation without relying on third-party services.



Figure 5.1: User Interface Overview Of The Algorithmic Resume Screening And Ranking System.

1.2 Functional Results (Application-Level)

The developed system features a convenient and easy-to-use web interface available to both individuals and recruiters. The dashboard acts as a central point where these two functionalities are distinguished explicitly, allowing users to switch between them without any difficulty.

In the individual screening mode, users are able to provide both their resume and the job description in order to obtain an extended evaluation of their profile. As a result, they get their total score calculated out of 100. Moreover, the system is capable of showing two additional scores – the semantic score and the keywords score, allowing users to understand how their total score was generated. One of the key benefits is the ability to see both matching and mismatching skills, which helps to determine the strengths and weaknesses of the applicant. Furthermore, if users choose the option of adding AI in their analysis, they will get some additional feedback. However, even without it, the core functionality will not be affected in any way.

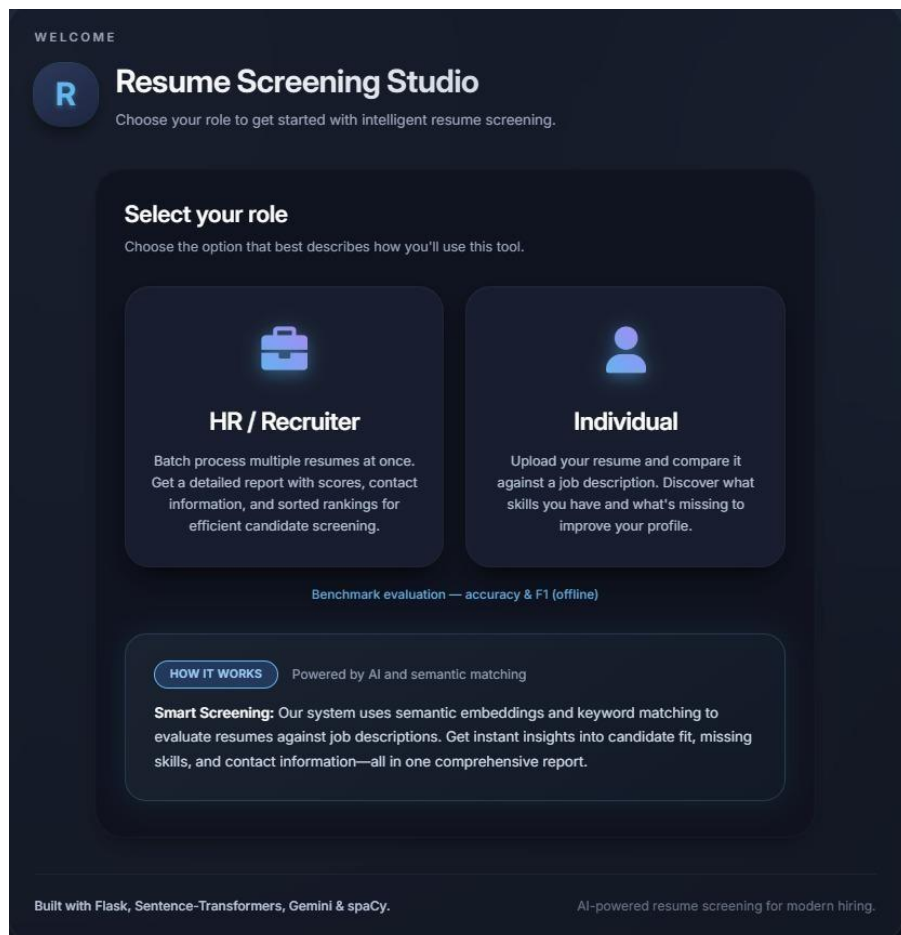


Fig. 5.2 Dashboard of SmartHire Application

In HR mode, recruiters are allowed to upload a set of different resumes for the one specific job description. The process works the same way as in the case of one file, except the system presents results in a certain ranked order, making the process of selecting the most appropriate applicants much easier. Another important benefit is that any error during file uploading does not prevent the system from processing other resumes.

An additional vital function is the extraction of the contact information (e.g., emails and telephone numbers) provided in resumes automatically. The mentioned process increases the operational effectiveness as there is no need for the manual search. Yet, it should be noted that the quality of the extraction can depend on the formatting of the resume. There are possibilities for a failure to provide accurate data for non-conforming resumes or even for those presented in an image format.

The developed tool provides an interface for evaluating the performance based on a specified sample. The information is presented through statistics and specific cases.

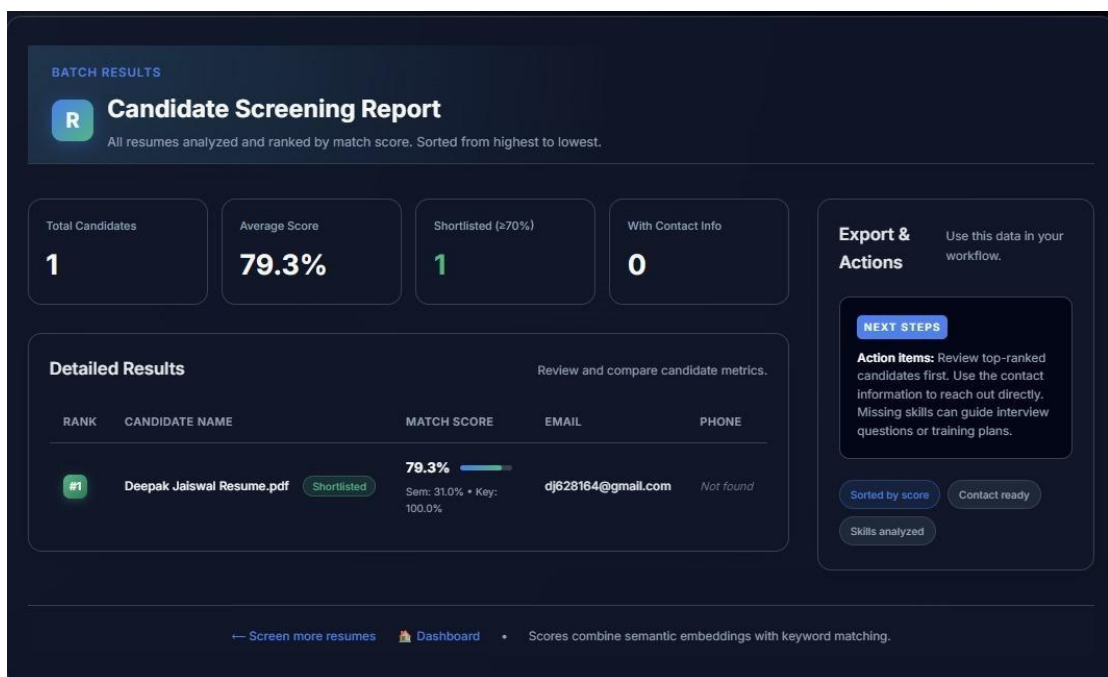


Fig. 5.3 Individual Based Analysis Of Resume Using SmartHire

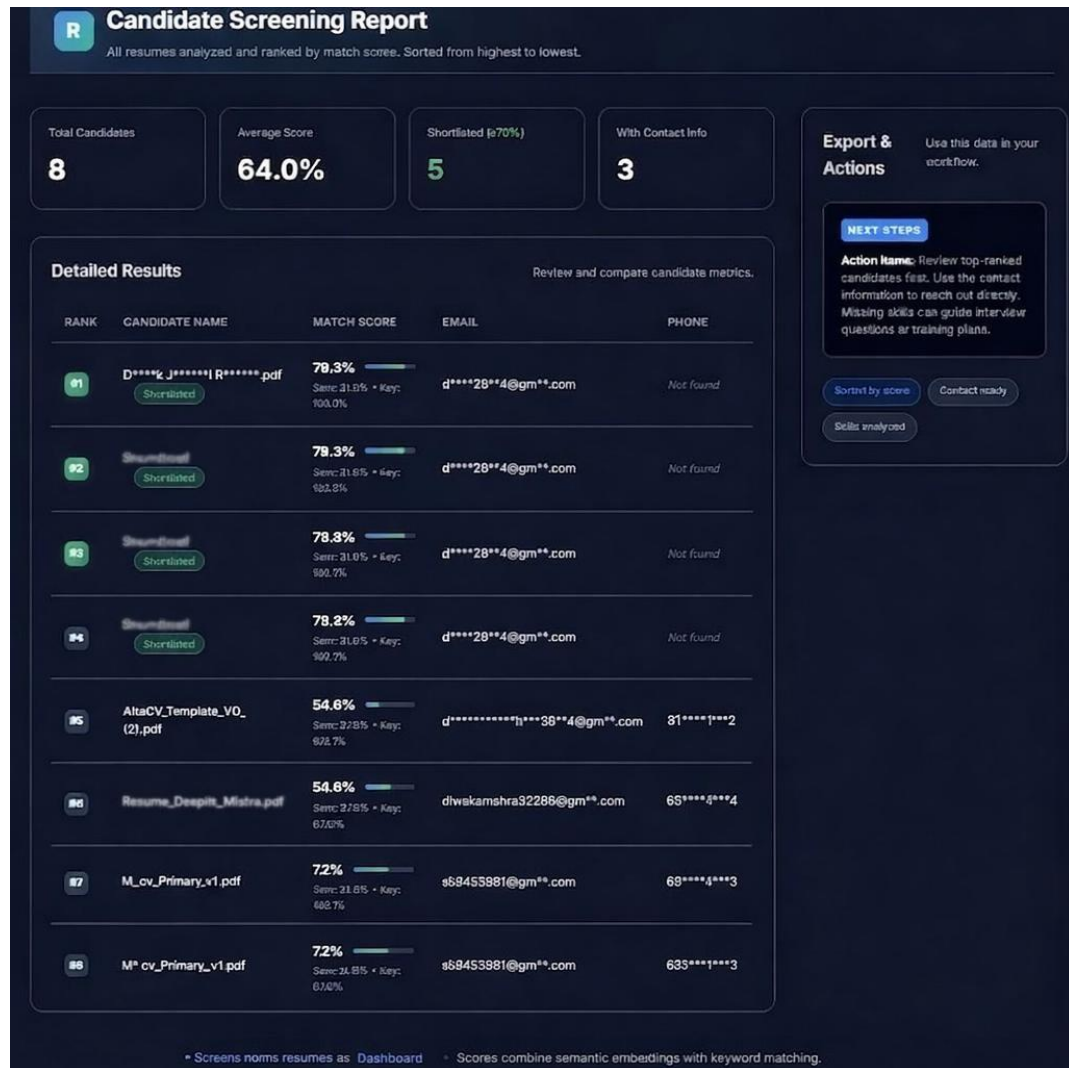


Fig. 5.4 HR/Recruiter Based Analysis Of Resume Using SmartHire

1.3 Results of Offline Quantitative Evaluation

Quantitative evaluation of the system is conducted based on labeled resume-job description pairs. In other words, each pair receives a ground truth label that reflects its relevance to a certain job. After that, the system processes a given pair in the same way, generating a particular score. Using a predetermined threshold, this score is translated into a binary value. Such an approach facilitates performance evaluation in terms of various well-known evaluation measures.

Evaluation of the results is made on the basis of such criteria as accuracy, precision, recall, and F1-score. These four evaluation measures are popular in the fields of classification and information retrieval [8]. They reflect different aspects of performance evaluation.

Analyzing the obtained results, it seems that associating semantic similarity with keyword

matching is a good approach.

Efficiency is increased by making use of the advantages of both methods. Semantic similarity helps to find relevant candidates even if different terms are used for the description. In addition, the criteria that are relevant for a search request can be detected by the method of keyword matching that the semantic similarity method tends to miss. The choice of the threshold value has a significant influence on the system's behavior. A high threshold value leads to a more selective selection of relevant candidates. A low threshold value makes it possible to find more relevant candidates.

We stress that the achieved performance metrics are for the specific dataset used for our evaluation, which may not generalize to larger or real world datasets. The training sets used for testing were smaller or artificially generated, which may overstate our model's capabilities.

1.4 Discussion

It is clear from the functional and quantitative evaluations performed that the proposed SmartHire system has been successful in accomplishing its desired purpose. The system has efficiently incorporated a pairwise resume-job matching process, provided an efficient hybrid score, and produced output that is easily understandable for all concerned parties. The dual functionality of the system is helpful in benefiting not only the job seeker but also the recruiters. Moreover, the evaluation module included in the system helps in performing the assessment.

One of the significant advantages of the proposed system is that it is capable of capturing the meaning of the context using the technique of semantic embedding. However, it is also possible to include the method of keyword analysis in order to verify the requirements. However, there are also some limitations that need to be mentioned. The efficiency of the system strongly depends on the variety of the evaluation data set, and the current system might be insufficient to take into account the more complicated features such as experience or interpersonal skills. The system lacks detection and mitigation of potential biases in the hiring process – another essential requirement for the automated hiring system [11], [12]. Also, scanning of non-text documents and images cannot be considered at this stage.

There are also some academic concerns regarding the study. The combined result is based only on textual similarity, and therefore, cannot reflect the actual suitability of the candidates. Moreover, the limited number of test samples might influence the generalization ability of the

results. Finally, usage of the threshold approach adds dependency on parameter choice. Summing up, we can say that while the developed SmartHire system lays a good foundation for intelligent resume screening, it needs further work to become a universal solution.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

The main idea behind the current project involved dealing with a growing problem of manual resume screening. For that reason, a web application of an intelligent resume screening system named SmartHire was proposed and implemented. The system makes use of a pairwise similarity approach, which involves calculating the match between a particular candidate's resume and a certain job role description. The resumes can be provided either in PDF or DOCX formats and are processed by extracting the text and using Natural Language Processing. The assessment process includes the use of a hybrid method, which incorporates the calculation of semantic similarity through sentence-transformer embedding and cosine similarity, together with keyword and domain-based matching of skills.

The major strength of the proposed system is the capability to generate not only one general score but also interpretable outputs as well. The system generates the semantic similarity, keyword matches, and missing skills, providing thus the possibility of obtaining a better insight regarding the obtained score.

The development involves two specific use cases using a Flask-based web interface. The first one is individual analysis where the user can upload his/her resume, get results, and receive feedback. The second one is called HR analysis, allowing the user to select several resumes in order to get a ranking list according to the relevance to a selected position. Other auxiliary tools like automatic extraction of email addresses and phones make the solution even more useful when dealing with resumes. Moreover, the option of using Google Gemini API for job processing and feedback generation is provided while the main algorithm works independently from third-party services.

In order to perform performance validation, an offline module is added to test the system on the basis of annotated data and measure such parameters as accuracy, precision, recall, and F1

score. Thus, all performance measures have been estimated in a systematic way. Overall, the project was successful, and all initial goals were achieved. The solution provides a practical and transparent way of dealing with resumes. It is possible to say that the use of semantic understanding along with keyword matching leads to better results.

6.2 Future Scope

While the present system provides a strong framework for intelligent resume filtering, some enhancements can still be achieved in many aspects of it. Regarding data collection and evaluation techniques, more robust results can be obtained through the utilization of larger and more varied datasets consisting of pairs of job positions and resumes. In addition, multi-annotator labeling for evaluation will increase the reliability of results. Finally, more sophisticated evaluation techniques like precision-recall curves and threshold analysis should be considered for the purpose of obtaining comprehensive insights about the system performance.

As far as the technical aspect of development is concerned, the system can be expanded in order to include structured features such as years of experience, educational qualification and other relevant credentials, and multilingual support for resumes by implementing advanced language models. OCR functionality may be considered in order to facilitate dealing with scanned resumes, whereas research related to fine-tuning embedding models on particular datasets should not be overlooked.

Regarding deployment and usability issues, authentication and authorization functionality may be implemented, along with the ability to integrate the tool with existing recruitment systems or ATS systems.

A second area where there are future possibilities for research involves issues such as ethics and fairness. Incorporating tools to identify bias and measure fairness to guard against influences could be a consideration in the implementation of this system. Additionally, adequate documentation should be provided to utilize the tool ethically and effectively.

6.3 Concluding Remarks

The SmartHire project has shown that state-of-the-art NLP methods as well as approaches using semantic similarity and simple keyword search can be put to effective use for finding the best candidate for a given job vacancy.

Automating and Facilitating the Resume Screening Process for Academic and Industrial

Purposes: enhance the efficiency and accountability of resume screening. Through the creation of an easy-to-use web application, we present a promising solution for both academic and industrial purposes.

This project provides the basis for further research and development on intelligent hiring platforms.

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